SUSCEPTIBILITY TO POSTBIOTIC SUBSTANCES OF ENTEROCOCCI ISOLATED FROM EWES' MILK LUMP CHEESES

EVA BINO, NATÁLIA ZÁBOLYOVÁ, MONIKA POGÁNY SIMONOVÁ, JANA ŠČERBOVÁ, ANDREA LAUKOVÁ

CENTRE OF BIOSCIENCES SLOVAK ACADEMY OF SCIENCES , INSTITUTE OF ANIMAL PHYSIOLOGY, KOSICE, SLOVAKIA

Ewes' milk represents 1.3% of global milk production and is widely used for traditional cheese making in Europe. Ewe's milk lump cheese is a traditional Slovak product made from unpasteurized milk without added starter cultures. Enterococci are common in raw ewe's milk and cheeses. They may occur naturally or as contaminants.

While enterococci contribute to fermentation and can produce beneficial postbiotic substances and bacteriocins, some strains are associated with infections. However, milk-derived enterococci generally lack major virulence factors compared to clinical strains.

POSTBIOTIC

Postbiotics are proteinaceous, non-homogeneously purified antimicrobial substances with inhibitory activity against related bacteria. When purified to homogeneity, they are classified as bacteriocins.

In this study, eight postbiotic substances were tested:

- 2 produced by *Lactococcus lactis* strains (MK2/8, MK1/3)
- 6 produced by *Enterococcus faecium* and *E. durans* strains (PS412, PS55, PS2019, PSED26E/7)
- 2 purified bacteriocins (Ent M, Ent A/P)

MATERIAL AND METHODS

- 31 isolates from 38 cheeses from Dairy Research Institute, Žilina (Slovakia)
  - *E. faecium (7)*
  - *E. faecalis* (19)
  - *Enterococcus* spp. (5)
- Agar spot test
- 8 postbiotic substances tested (our laboratory)

Activity expressed as AU/ml (highest dilution inhibiting growth).

Positive control: *Enterococcus avium* EA5.

All experiments were performed in duplicate.

RESULTS AND DISCUSSION

Postbiotic substances showed variable inhibitory activity against 31 enterococci isolated from ewe's milk lump cheeses.

Among *Lactococcus lactis* postbiotics, MK2/8 inhibited 52% of strains and MK1/3 inhibited 71%, both reaching up to 1,600 AU/ml.

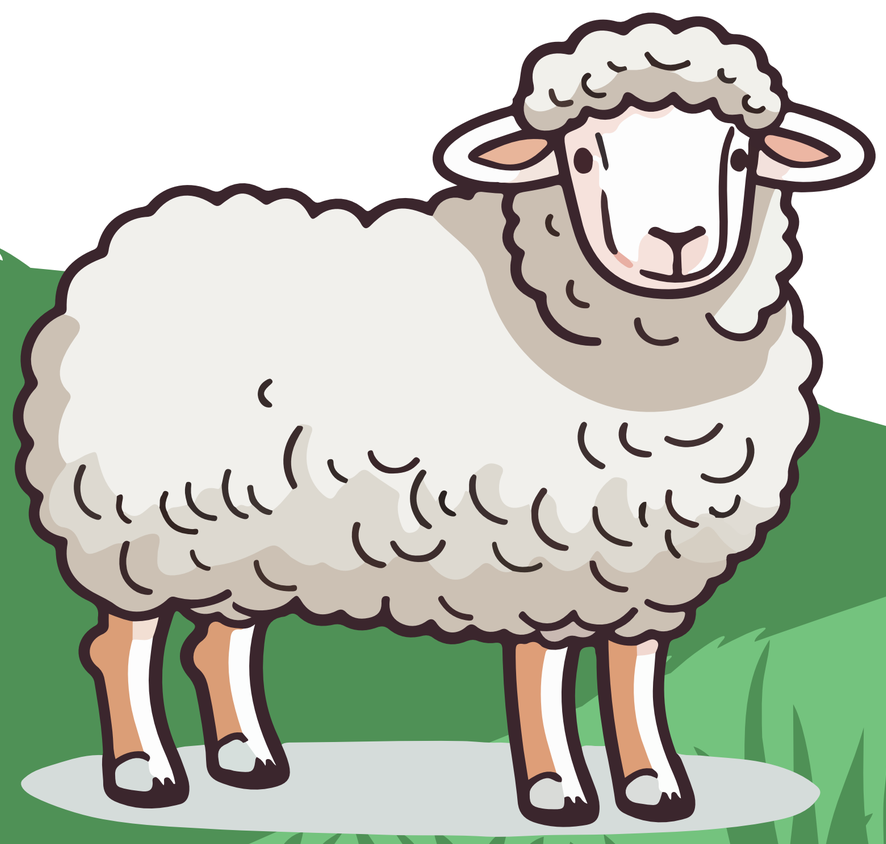
The most effective were PS produced by enterococci: PS412 and PS2019, which inhibited 94% of strains with activity up to 3,200 AU/ml. PSEF55 also showed strong efficacy (91%, 3,200 AU/ml). Ent M inhibited 71% of strains and EK13 74%. The lowest activity was observed for ED26E/7 (55% inhibition; max 100 AU/ml).

E. faecium isolates were susceptible to almost all postbiotics, with rare resistance to Ent A/P.

Enterococcus spp. showed limited resistance mainly to Ent A/P.

Among *E. faecalis*, high susceptibility was observed toward PS412, PS55, PS2019, Ent A/P, and Ent M, whereas MK2/8 (32%), MK1/3 (58%), and especially ED26E/7 (26%) were less effective.

These findings confirm the strong in vitro antimicrobial potential of selected enterococcal postbiotics, particularly PS412 and PS2019, suggesting their promising application in improving microbiological safety of traditional ewe's milk products.



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